

Partial traces in orthonormal coordinates

Sirui Lu Erickson Tjoa J. Ignacio Cirac

Last update: October 8, 2026

Contents

1	Partial Traces in Orthonormal Coordinates	2
---	---	---

Chapter 1

Partial Traces in Orthonormal Coordinates

We record the coordinate and integrability facts used to estimate physical output operators in [Ope26, proof of Theorem 5.2], 04-compression.tex, lines 383–549. The matrices need not be positive or self-adjoint. All norms denoted by $\|\cdot\|_1$ are rectangular trace norms.

Theorem 1.1 (Isometric coordinates and the reduced trace norm). [Lean](#) [✓ Stmt](#) *Let X, Y, D, E be finite index sets and let $U : \mathbb{C}^X \rightarrow \mathbb{C}^Y$ and $V : \mathbb{C}^D \rightarrow \mathbb{C}^E$ satisfy $U^\dagger U = I$ and $V^\dagger V = I$. Every matrix M on $\mathbb{C}^X \otimes \mathbb{C}^D$ satisfies*

$$\|\mathrm{tr}_E((U \otimes V)M(U \otimes V)^\dagger)\|_1 = \|\mathrm{tr}_D M\|_1. \quad (1.1)$$

Neither isometry is required to be onto.

Proof. [✓ Proof](#) The partial trace on the left equals $U(\mathrm{tr}_D M)U^\dagger$ because $V^\dagger V = I$. Left and right multiplication by isometries preserve the nonzero singular values, giving (1.1). $\square$

Theorem 1.2 (Independence of product orthonormal coordinates). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{H}, \mathcal{K}$ be complex inner-product spaces with finite orthonormal bases b_X, b_Y and b_D, b_E , respectively. Write C_{YX} and C_{ED} for the two changes of coordinates. The change from $b_X \otimes b_D$ to $b_Y \otimes b_E$ is $C_{YX} \otimes C_{ED}$. If $[z]_{XD}$ denotes product coordinates, then for arbitrary $z, t \in \mathcal{H} \otimes \mathcal{K}$,*

$$(C_{YX} \otimes C_{ED})[z]_{XD}[t]_{XD}^\dagger (C_{YX} \otimes C_{ED})^\dagger = [z]_{YE}[t]_{YE}^\dagger. \quad (1.2)$$

Consequently, for every finite family z_i, t_i and complex coefficients c_i ,

$$\left\| \mathrm{tr}_E \sum_i c_i [z_i]_{YE} [t_i]_{YE}^\dagger \right\|_1 = \left\| \mathrm{tr}_D \sum_i c_i [z_i]_{XD} [t_i]_{XD}^\dagger \right\|_1. \quad (1.3)$$

Proof. [✓ Proof](#) A product-basis coordinate is the product of the two endpoint coordinates, which gives the Kronecker formula for the change of basis. Multiplication by that matrix sends $[z]_{XD}$ to $[z]_{YE}$, proving (1.2). Distribute the coordinate changes through the finite sum and apply Theorem 1.1. $\square$

Theorem 1.3 (Reconstruction from rectangular blocks). [Lean](#) [Lean](#) [✓ Stmt](#) Let M have rows indexed by $X \times I$ and columns indexed by $Y \times J$, where X, Y are finite. Put $(M_{xy})_{ij} = M_{(x,i),(y,j)}$. Then

$$M = \sum_{x \in X} \sum_{y \in Y} |x\rangle\langle y| \otimes M_{xy}. \quad (1.4)$$

If I, J are also finite, then

$$\|M\|_1 \leq \sum_{x \in X} \sum_{y \in Y} \|M_{xy}\|_1. \quad (1.5)$$

No factor depending on the dimensions of the blocks appears.

Proof. [✓ Proof](#) At entry $((x, i), (y, j))$, all matrix-unit summands except (x, y) vanish. For the inequality, apply the triangle inequality to (1.4) and use $\| |x\rangle\langle y| \otimes N \|_1 = \|N\|_1$. $\square$

Theorem 1.4 (Blocks commute with the discarded partial trace). [Lean](#) [✓ Stmt](#) Let X, Y be arbitrary index sets, let D be finite, and let M be a matrix with rows and columns indexed by $(X \times Y) \times D$. Fix $x, x' \in X$. Taking the (x, x') block after tracing out D gives the same matrix as first taking that block and then tracing out D :

$$(\mathrm{tr}_D M)_{xx'} = \mathrm{tr}_D (M_{xx'}). \quad (1.6)$$

Proof. [✓ Proof](#) Both sides have entry $\sum_{d \in D} M_{((x,y),d),((x',y'),d)}$ at (y, y') . $\square$

Theorem 1.5 (An averaged block bound). [Lean](#) [✓ Stmt](#) Let (Ω, μ) be a measure space and let $M(\omega)$ be a finite rectangular block matrix as in Theorem 1.3. Suppose every function $\omega \mapsto \|M_{xy}(\omega)\|_1$ is integrable. Then

$$\int \|M(\omega)\|_1 d\mu(\omega) \leq \sum_{x,y} \int \|M_{xy}(\omega)\|_1 d\mu(\omega). \quad (1.7)$$

Here the Bochner integral is assigned the value zero when its integrand is not integrable. The assertion does not itself imply integrability of the full trace norm.

Proof. [✓ Proof](#) The sum of the block norms is integrable and dominates the nonnegative full trace norm by (1.5). If the latter is integrable, integrate this inequality and exchange the finite sum with the integral. Otherwise the left side is zero by the stated convention, whereas the right side is nonnegative. $\square$

Theorem 1.6 (Integrability of finite matrix combinations). [Lean](#) [Lean](#) [✓ Stmt](#) Let μ be a finite measure, let J be finite, and fix finite rectangular complex matrices A_j . Suppose the scalar functions $z_j : \Omega \rightarrow \mathbb{C}$ satisfy $z_i \bar{z}_j \in L^1(\mu)$ for all $i, j \in J$. Then $\omega \mapsto \|\sum_j z_j(\omega) A_j\|_1$ is integrable. In particular, it is enough that each z_j belongs to $L^2(\mu)$. No independence or prescribed covariance is assumed.

Proof. [✓ Proof](#) The diagonal product hypothesis gives $|z_j|^2 \in L^1$. Taking its square root gives measurability of $|z_j|$, and finiteness of the measure then gives $|z_j| \in L^1$. The Gram matrix of $\sum_j z_j A_j$ depends only on the measurable pair products. Its positive square root therefore shows that the trace norm is measurable. The pointwise bound

$$\left\| \sum_j z_j A_j \right\|_1 \leq \sum_j |z_j| \|A_j\|_1 \quad (1.8)$$

gives integrability by domination. Finally, if all coefficients are square-integrable, complex conjugation preserves that property and Cauchy–Schwarz gives the required pairwise integrability. $\square$

Bibliography

- [Ope26] OpenAI. Polynomial peps approximation of gapped square-grid ground states, 2026. September 24, 2026; source revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.